

(+91) 70692-24818
New Delhi, India
pateltanvi325@gmail.com

Tanvi Patel

github.com/Tanu1007
linkedin.com/in/tanvipatel1007

To pursue advanced work in Physical AI and embodied agents by integrating robust perception, efficient reasoning, and real-time system design. I seek opportunities where I can apply my background in computer vision, embedded AI, and scalable infrastructure to develop intelligent, interactive autonomous systems.

Area of Expertise: Backend Development, Microservices Architecture, Distributed Systems, Cloud Computing, Machine Learning, Problem Solving, Data Structures & Algorithms.

EDUCATION

B.Tech – National Institute of Technology, Surat
Electronics and Communication Engineering

Aug 2018 – May 2022
8.6/10

PROJECTS

Self-powered IoT Devices: An Application of Indoor Photovoltaics 🔗

Jan 2022 – May 2022

Technologies Used: SCAPS-1D, WebPlotDigitizer, MATLAB, Device Simulation, Data Analysis

- Simulated and optimized a **lead-free perovskite solar cell (Cs₂AgBiBr₆)** achieving up to **28.9% efficiency** under white LED illumination for indoor energy harvesting.
- Proposed a **sustainable, non-toxic photovoltaic solution** to power low-energy **IoT devices and sensors** through indoor light harvesting.

Localization using Optical Camera Communication (OCC)

Jan 2022 – May 2022

Technologies Used: MATLAB, OpenCV, Image Processing, Photogrammetry, Computer Vision

- Developed a vision-based vehicle localization framework integrating **OCC** and **photogrammetry** to estimate positions using LED light signals.
- Implemented distance and angle estimation models for **forward vehicles** and **streetlights**, enabling real-time relative position tracking of the host vehicle.

Indian Rover Challenge (IRC 2020) 🔗

Aug 2019 – Jan 2020

Technologies Used: Python, Keras, TensorFlow, CNN, OpenCV

- Developed a computer vision module for a Mars rover prototype using CNNs for **terrain navigation** and object detection.
- Achieved over **92%** accuracy in terrain segmentation and implemented real-time obstacle detection with OpenCV for autonomous navigation.

Swadeshi Microprocessor Challenge

Sep 2020 – Jan 2021

Technologies Used: Embedded Deep Learning, CNN, Thermal Image Processing

- Designed and deployed an **embedded CNN model** for real-time thermal image classification with 94% detection accuracy.
- Optimized neural network inference for limited-resource microprocessor hardware, improving throughput by **38%** without compromising model precision.

TECHNICAL EXPERIENCE

Airtel X Labs

Jul 2022 – Oct 2025

Backend Engineer (Deputy Manager)

Gurugram, Haryana

Technologies Used: Java, Spring Boot, Kafka, Aerospike, Elasticsearch, MySQL, MongoDB, Grafana, FluentD, StatsD, Jolokia

Airtel CPaaS International:

- Led deployments across **10 OpCos** (Kenya, Tanzania, DRC, etc.), enhancing communication services for millions of users.
- Engineered **Prepaid APIs** and migrated the wallet system from SQL to Aerospike, achieving **5x faster transactions** and supporting **1000+ TPS**.
- Integrated **Payment Gateway modules** (Airtel Money, DPO, Cellulant) with robust callback and refund mechanisms.

Airtel IQ Platform Team

- Implemented **2FA and OAuth 2.0** for Airtel IQ Dashboard, improving authentication speed by **15%**.
- Built the **IQ Contact Book microservice** for 1000+ enterprise clients supporting bulk upserts, segmentation, and campaign scheduling.
- Automated **SMS onboarding and SIM call-recording workflows** for the System Integration (SI) team, eliminating repetitive manual tasks and improving process efficiency by **60%** through backend-driven orchestration and monitoring.

Airtel IQ SMS Team

- **Rearchitected Messaging Service**, increasing throughput from **1500 TPS to 3500 TPS** by eliminating Kafka connector bottlenecks and optimizing event flow.

Wipro Limited

Jun 2021 – Jul 2021

Turbo Intern

Remote

- Developed a real-time **computer vision system** using Python and OpenCV to analyze ECG graphs, deployed across **100+ ICUs** to enhance critical patient monitoring.
- Improved medical data accuracy and response efficiency through automated anomaly detection and real-time alerting.

Geliose Mobility

Jun 2021 – Jul 2021

Junior Mobile App Developer

Remote

- Built an **Android app for an EV startup** featuring user authentication, EV mapping, Bluetooth discovery, and real-time chat, streamlining communication for **5,000+ users** and enhancing operational efficiency.

SKILLS

Programming Languages	Java, Python, C/C++, SQL
Frameworks	Spring Boot, Spring MVC, Kafka, RabbitMQ, AWS SQS, Keras, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV
Databases	MongoDB, Elasticsearch, Aerospike, Redis, AWS, S3, Cassandra, GCP, MySQL
Platforms	Git/Github, Grafana, Kong, Jenkins, JIRA, Android Studio
Programming Paradigms	Microservices Architecture, Reactive and Event-Driven Programming, AOP, LLD, DSA, Software Design Patterns, Machine Learning, Deep Learning, Computer Vision

ACHIEVEMENTS & CO-CURRICULAR

- Secured **3rd position** in **Airtel's Hackathon (Samadhan)** for implementing a solution to reduce OTP-based frauds and enhance transaction security.
- Achieved **9th** rank in **Indian Rover Challenge (IRC 2020)**, **13th** rank in **International Rover Design Challenge (IRDC 2020)**, and **17th** rank in **International Mars Hackathon (IMH 2020)** organized by **MSSA**.
- **Training and Placement Coordinator, ECE Department** : Coordinated campus placement activities for 200+ students, liaised with recruiters, and facilitated interview readiness programs.
- **Mentor, Drishti Technical Club** : Guided junior members on machine learning projects; conducted hands-on workshops and project reviews.

ADDITIONAL COURSES

• Spring MVC, Spring Boot and REST Controllers 	2022
• Spring – Ecosystem and Core 	2022
• Java Programming: Solving Problems with Software 	2022
• Neural Networks and Deep Learning 	2020